

1 Strongly Rayleigh Distributions

We have already defined real stable polynomials and demonstrated a number of closure properties, for example partial differentiation and specialization. We also talked about Borcea and Brändén's characterization of differential operators.

In the first lecture, we characterized the univariate distributions which had real rooted generating polynomials. In this lecture we more formally define and study the distributions whose generating polynomials are real stable.

Recall that given a distribution $\mu : \mathbb{Z}_{\geq 0}^n \rightarrow [0, 1]$ with finite support, its generating polynomial is:

$$p_\mu(z_1, \dots, z_n) = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^n} \mu(\alpha) z^\alpha$$

Definition 1.1 (Strongly Rayleigh). *A distribution $\mu : \mathbb{Z}_{\geq 0}^n \rightarrow [0, 1]$ is strongly Rayleigh if p_μ is real stable.*

1.1 Closure Properties

One of the main reasons the strong Rayleigh (SR) property is useful is that if μ is SR, so are many distributions that arise from μ . Suppose μ is SR. Then, the following are also SR (although there are many more closure properties):

1. **Conditioning.** $\mu|_{i=0}$ is SR, as its generating polynomial is a scaling of $p_\mu|_{z_i=0}$. Recall for an event S , $\mu|_S$ is the distribution μ conditioned on S , so here we are conditioning i "out." This assumes $\mathbb{P}[i = 0] > 0$.

Similarly, if the maximum degree of z_i is 1, $\mu|_{i=1}$ is SR, as its generating polynomial is a scaling of $z_i \partial_{z_i} p_\mu$ (in other words, we condition i "in", and again here we need $\mathbb{P}[i = 1] > 0$). This is stable, as the operator $z_i \partial_{z_i}$ is associated to $-z_i w_i$, which is a stable polynomial.

2. **Projection.** Given a set $F \subseteq [n]$, we can project the distribution to the set F , which we write $\mu|_F$. In other words, for each element of the support we ignore the elements not in F . This is:

$$\mu|_F(\alpha) = \sum_{\beta \in \mathbb{Z}_{\geq 0}^n : \beta_F = \alpha} \mu(\beta)$$

and the generating polynomial of this is simply $p_\mu|_{z_i=1 \forall i \notin F}$, which is stable by specialization.

3. **External Field.** We can apply external fields. For $\lambda \in \mathbb{R}_{\geq 0}^n$, $\mu_\lambda(\alpha) = \mu(\alpha) \lambda^\alpha$, up to normalization. The generating polynomial here is $p_\mu(\lambda_1 z_1, \dots, \lambda_n z_n)$ up to normalization.

2 Random Spanning Trees

Consider the uniform distribution over spanning trees of a graph $G = (V, E)$. If $\mathcal{T}$ is the set of all trees of G , this has generating polynomial:

$$p_\mu(\{z_e\}_{e \in E}) = \frac{\sum_{T \in \mathcal{T}} \prod_{e \in T} z_e}{|\mathcal{T}|}$$

In this lecture, we will prove that the uniform distribution over trees is SR, i.e., that this polynomial is real stable for any graph G . It will follow by external field that λ -uniform distributions are SR as well, where a λ -uniform distribution is given by the graph and a vector $\lambda \in \mathbb{R}_{\geq 0}^{|E|}$ which has generating polynomial:

$$p_{\mu_\lambda}(\{z_e\}_{e \in E}) = \frac{\sum_{T \in \mathcal{T}} \prod_{e \in T} \lambda_e z_e}{\sum_{T \in \mathcal{T}} \prod_{e \in T} \lambda_e}$$

in other words, μ_λ as defined above.

To prove these distributions are SR, we first introduce the following lemma which generates many interesting examples of strong Rayleigh distributions. Notice the following consequence of the fact that λ -uniform distributions are SR. Let $S \subseteq V$. Then the number of edges sampled in $\delta(S)$ is distributed as a Poisson binomial. This is a strong and surprising fact, because the edges are not independent, yet the number of them that appear is modeled by independent Bernoullis.

2.1 Brief Linear Algebra Refresher

Recall that a symmetric matrix is positive semidefinite (PSD) if all of its eigenvalues are non-negative, and positive definite (PD) if all of its eigenvalues are positive.

Also recall that by the spectral theorem, any symmetric matrix $M \in \mathbb{R}^{n \times n}$ with eigenvalues $\gamma_1, \dots, \gamma_n$ with orthonormal eigenvectors $v_1, \dots, v_n$ can be written $M = \sum_{i=1}^n \gamma_i v_i v_i^T$. Thus, a PD matrix always has a square root $M^{1/2} = \sum_{i=1}^n \sqrt{\gamma_i} v_i v_i^T$ and similarly $M^{-1/2} = \sum_{i=1}^n \gamma_i^{-1/2} v_i v_i^T$.

Finally, we will use that the characteristic polynomial of a square matrix $M \in \mathbb{R}^{n \times n}$ is the univariate polynomial $\det(xI - M)$, whose roots are the eigenvalues of M . As a warm up, let's recall why this is the case. If x is a root of $\det(xI - M)$, then $\det(xI - M) = 0$. It follows that $xI - M$ is a singular matrix, so there is a non-zero vector v so that $(xI - M)v = 0$, i.e., $xIv - Mv = 0$ or $xv = Mv$.

A key fact we will use is that if M is symmetric, its eigenvalues are real, so $\det(xI - M)$ is real rooted. Let's recall why this is the case too. Let v be an eigenvector of M with eigenvalue γ . Then:

$$\gamma \|v\|^2 = v^* \gamma v = v^* M v$$

this is real, since $\overline{v^* M v} = v^* M^* v = v^* M v$. Since $\|v\|^2 \neq 0$, γ must be real.

2.2 Mother of All SR Distributions

To prove that λ -uniform spanning tree distributions are SR, we will use the following lemma, sometimes called the "mother of all" SR distributions.

Lemma 2.1. Suppose $p(z_1, \dots, z_n) = \det(\sum_{i=1}^n z_i P_i + S)$ for PSD matrices $P_1, \dots, P_n \in \mathbb{R}^{d \times d}$ and a symmetric matrix $S \in \mathbb{R}^{d \times d}$. Then, p is real stable (or identically zero).

Proof. Consider any $a \in \mathbb{R}_{>0}^n, b \in \mathbb{R}^n$ and the polynomial

$$q(x) = \det \left(\sum_{i=1}^n (a_i x + b_i) P_i + S \right) = \det \left(x \sum_{i=1}^n a_i P_i + \sum_{i=1}^n b_i P_i + S \right).$$

By the line test, it is enough to prove that q is real rooted for all choices a, b .

First assume that each P_i is positive definite. Then $D = \sum_{i=1}^n a_i P_i$ is positive definite since $a \in \mathbb{R}_{>0}^n$. Therefore we can rewrite this polynomial as

$$\det(D^{1/2}) \det \left(xI + D^{-1/2} \left(\sum_{i=1}^n b_i P_i + S \right) D^{-1/2} \right) \det(D^{1/2})$$

$\det(D^{1/2})$ is a constant, so this is real rooted if $\det(xI + D^{-1/2}(\sum_{i=1}^n b_i P_i + S)D^{-1/2})$ is real rooted. However this is the characteristic polynomial of the matrix $Q = -D^{-1/2}(\sum_{i=1}^n b_i P_i + S)D^{-1/2}$, so its roots are the eigenvalues of Q . Finally we observe that Q is symmetric since all P_i and S are symmetric, so its eigenvalues are all real, implying that this polynomial is real rooted as desired.

If the P_i are not positive definite, this follows from taking the limit of $P_i + \epsilon I$ as $\epsilon \rightarrow 0$ and applying Hurwitz. $\square$

2.3 Matrix Tree Theorem

As you might suspect, λ -uniform spanning tree distributions can be generated from the above, by Kirchhoff's matrix tree theorem. We will use the following:

Lemma 2.2 (Cauchy-Binet). *Let $A \in \mathbb{R}^{n \times m}$. Then:*

$$\det(AA^T) = \sum_{S \in \binom{[m]}{n}} \det(A_S)^2$$

where A_S consists of the columns of A indexed by S .

Notice this is like a higher-dimensional pythagorean theorem. A consequence is that for a diagonal matrix $Z = \text{diag}(z_1, \dots, z_m)$, $\det(BZB^T)$ is, taking $A = BZ^{1/2}$:

$$\det(BZB^T) = \sum_{S \in \binom{[m]}{n}} \det((BZ^{1/2})_S)^2 = \sum_{S \in \binom{[m]}{n}} \det(B_S)^2 z^S$$

we can do this whenever all $z_i \geq 0$ so that the square root exists. But since the two polynomials agree on the positive orthant, they must agree everywhere: $p_1(z) - p_2(z)$ is zero on an open set, so it must be zero everywhere.

Now we can prove the theorem:

Theorem 2.3 (Matrix tree theorem). *For a graph $G = (V, E)$ let $g_{\mathcal{T}}(\{z_e\}_{e \in E}) = \sum_{T \in \mathcal{T}} \prod_{e \in T} z_e$ be the unnormalized generating polynomial of the 1-uniform spanning tree distribution on G . Pick a root $r \in V$ arbitrarily. Then, where $|V| = n$,*

$$g_{\mathcal{T}}(\{z_e\}_{e \in E}) = \det \left(\sum_{e \in E} z_e L_e \right),$$

where for $e = \{u, v\}$ with $u, v \neq r$ we let $L_e = (\mathbf{1}_u - \mathbf{1}_v)(\mathbf{1}_u - \mathbf{1}_v)^T \in \mathbb{R}^{(n-1) \times (n-1)}$ be the Laplacian of e . For $e = \{u, r\}$ we let $L_e = \mathbf{1}_u \mathbf{1}_u^T$.

Proof. Let $b_e = (\mathbf{1}_u - \mathbf{1}_v)$ for $e = \{u, v\}$ with $u, v \neq r$ and $b_e = \mathbf{1}_u$ for $e = \{r, u\}$. Let $B \in \mathbb{R}^{(n-1) \times m}$ be the matrix with columns b_e for all $e \in E$ (where $|E| = m$). Then, by Cauchy-Binet, where Z is the diagonal matrix with entries z_e ,

$$\det\left(\sum_{e \in E} z_e L_e\right) = \det(BZB^T) = \sum_{S \in \binom{[m]}{n-1}} z^S \det(B_S)^2$$

If the edges in S contain a cycle, then $\det(B_S) = 0$ as the set of edges in the cycle are linearly dependent. Otherwise, since $|S| = n - 1$ and is acyclic, it is a spanning tree. So it remains to show that the determinant is ± 1 if S forms a spanning tree. To see this, notice that since the tree contains a non-root leaf, there must be a row of the matrix with exactly one non-zero “leaf” entry. We can place this entry in the bottom right of the matrix, and recurse (considering the leaf as deleted in the tree and the matrix), leaving us with an upper triangular matrix. This implies $\det(B_S) = \pm 1$ as desired. $\square$

Sometimes you will see the following slightly more convenient expression:

$$g_{\mathcal{T}}(\{z_e\}_{e \in E}) = \frac{1}{n} \det\left(\sum_{e \in E} z_e L_e + \mathbf{1}\mathbf{1}^T/n\right) \quad (1)$$

where $L_e = (\mathbf{1}_u - \mathbf{1}_v)(\mathbf{1}_u - \mathbf{1}_v)^T$ for all $e = \{u, v\}$. We leave the proof of the equivalence as an exercise. Both forms show that the distribution is SR. In the version from the lemma we can choose the symmetric matrix to be all 0s, and in the form here $\mathbf{1}\mathbf{1}^T/n$ is symmetric. Furthermore, the matrices L_e are PSD since they are equal to vv^T for some v .